

Main Project 03 – Web Tour Applications

Introduction

In this Web Tour Applications, which is used to simulate flight booking on two cities with round trips. It contains a home page, itinerary page, flight booking page and sign off page. We are going to perform the load test, stress test, spike test, endurance test, and volume test here.

- Web application at <http://localhost:1080/Webtours/>

Test Setup

Weather Data API

- | | |
|--|--------------------|
| 1. http://localhost:1080/Webtours/cgi-bin/login.pl | - Login Page |
| 2. http://localhost:1080/Webtours/cgi-bin/nav.pl | - Flights Page |
| 3. http://localhost:1080/Webtours/cgi-bin/welcome.pl | - Welcome Page |
| 4. http://localhost:1080/Webtours/cgi-bin/reservations.pl | - Reservation Page |
| 5. http://localhost:1080/Webtours/cgi-bin/cgi-bin/welcome.pl | - Sign out Page |

Test-Setup

1. From blaze meter to get the data from website.
2. Initiate Web tours server in the local folder.
3. Using Synchronizing Timer to integrate no of user requests.
4. Using Random Controller to process the requests on randomly.
5. Using Listeners to get the data.
6. Using config elements to record used to provide default values when making a request to a server with samplers.

Environment Setup

1. **Processor** - 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 2.42 GHz
2. **RAM** – 8GB

Jenkins

It integrated with Jenkin's.

GitHub

1. https://github.com/senthilraja29/Performance_Testing/blob/master/Web-Tours%20-%20Endurance%20Test.jmx
2. https://github.com/senthilraja29/Performance_Testing/blob/master/Web-Tours%20-%20Flight%20Booking%20Stress%20Testing.jmx
3. https://github.com/senthilraja29/Performance_Testing/blob/master/Web-Tours%20-%20Itinerary%20Management%20Load%20Testing.jmx
4. https://github.com/senthilraja29/Performance_Testing/blob/master/Web-Tours%20-%20Sign%20Off%20Stress%20Testing.jmx
5. https://github.com/senthilraja29/Performance_Testing/blob/master/Web-Tours%20-%20Load%20Test.jmx.
6. https://github.com/senthilraja29/Performance_Testing/blob/master/Web-Tours%20-%20Sign%20Off%20Stress%20Testing.jmx
7. https://github.com/senthilraja29/Performance_Testing/blob/master/Web-Tours%20-%20Spike%20Test.jmx

8. https://github.com/senthilraja29/Performance_Testing/blob/master/Web-Tours%20-%20Volume%20Test.jmx

Environment

Running Environment Jenkins.

Test Results

Load Test

In this load test to perform 200 concurrent users hitting at 63 mints by using concurrency thread group. After conducting load test around 42.3 thousand users hit the website within 30 mints, we received **0.01%** of error.

Summary Report

Name:

Summary Report

Comments:

Write results to file / Read from file

Filename:

\Testing\performance testing\Projects\Main 03\Load Test\Web-Tours - Load Test.cs

Browse...

Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB...	Sent KB/sec	Avg. Bytes
Home Page	2379	10941	110	394898	19436.49	0.00%	37.8/min	1.08	0.32	1750.0
Login	4614	11016	160	394918	20489.30	0.02%	1.2/sec	1.31	0.73	1096.3
Researvation	2326	11061	182	394854	21201.53	0.00%	37.0/min	2.72	0.32	4521.0
Signout	2382	11184	190	394722	22314.42	0.00%	37.8/min	0.63	0.32	1026.0
Welcome	2342	11172	225	394729	22509.32	0.04%	37.2/min	0.51	0.32	843.9
Book ticket	2356	11488	221	394871	23773.77	0.00%	37.4/min	1.65	0.50	2711.0
Book seat	2359	11056	200	394702	21011.51	0.04%	37.5/min	1.82	0.45	2976.0
Itinerary Page	2382	10823	231	394402	19409.91	0.00%	37.8/min	0.71	0.32	1155.0
Itinerary Pag...	2369	10836	135	394511	19461.53	0.00%	37.6/min	0.50	0.32	819.0
researvation ...	2443	10686	176	394606	17535.27	0.00%	38.9/min	2.44	0.52	3858.0
researvation	2314	10514	186	399966	14184.69	0.00%	36.9/min	1.68	0.55	2792.0
confirmation	2376	10469	202	392858	13883.41	0.00%	37.9/min	1.77	0.58	2871.7
re-researvati...	2332	10656	173	394611	16098.95	0.00%	37.2/min	2.73	0.38	4521.0
researvation ...	2383	11038	249	394718	20927.17	0.00%	38.0/min	1.85	0.48	2995.6
Itinerary Pag...	2367	10555	224	394497	15963.71	0.00%	37.7/min	1.07	0.33	1750.0
Flights	2295	10562	160	394518	16244.43	0.00%	36.6/min	1.04	0.31	1750.0
Cancel ticket	2360	10561	272	394080	15982.91	0.00%	37.6/min	0.71	0.42	1155.0
TOTAL	42379	10869	110	399966	19161.13	0.01%	11.2/sec	24.19	7.16	2207.0

Aggregate Report

Name:

Aggregate Report

Comments:

Write results to file / Read from file

Filename

Browse...

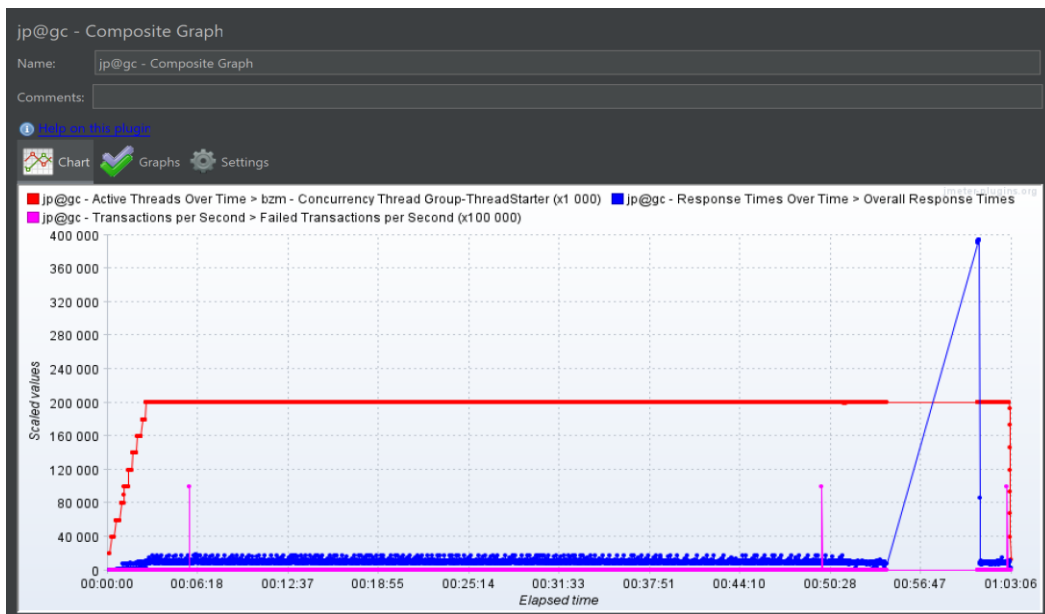
Log/Display Only:

☐ Errors

☐ Successes

Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughp...	Received ...	Sent KB/sec
Home Page	2379	10941	10399	11749	17478	17723	110	394898	0.00%	37.8/min	1.08	0.32
Login	4614	11016	10395	11695	17453	17678	160	394918	0.02%	1.2/sec	1.31	0.73
Researvati...	2326	11061	10403	11711	17455	17692	182	394854	0.00%	37.0/min	2.72	0.32
Signout	2382	11184	10416	11633	17421	17721	190	394722	0.00%	37.8/min	0.63	0.32
Welcome	2342	11172	10373	11656	17441	17690	225	394729	0.04%	37.2/min	0.51	0.32
Book ticket	2356	11488	10511	11745	17483	17732	221	394871	0.00%	37.4/min	1.65	0.50
Book seat	2359	11056	10391	11689	17458	17717	200	394702	0.04%	37.5/min	1.82	0.45
Itinerary P...	2382	10823	10376	11680	17411	17662	231	394402	0.00%	37.8/min	0.71	0.32
Itinerary P...	2369	10836	10379	11628	17407	17645	135	394511	0.00%	37.6/min	0.50	0.32
researvati...	2443	10686	10399	11700	17434	17664	176	394606	0.00%	38.9/min	2.44	0.52
researvati...	2314	10514	10445	11830	17503	17684	186	399966	0.00%	36.9/min	1.68	0.55
confirmati...	2376	10469	10414	11815	17489	17707	202	392858	0.00%	37.9/min	1.77	0.58
re-researv...	2332	10656	10439	11804	17458	17717	173	394611	0.00%	37.2/min	2.73	0.38
researvati...	2383	11038	10433	11760	17438	17692	249	394718	0.00%	38.0/min	1.85	0.48
Itinerary P...	2367	10555	10373	11731	17485	17665	224	394497	0.00%	37.7/min	1.07	0.33
Flights	2295	10562	10359	11655	17445	17737	160	394518	0.00%	36.6/min	1.04	0.31
Cancel tic...	2360	10561	10374	11731	17454	17691	272	394080	0.00%	37.6/min	0.71	0.42
TOTAL	42379	10869	10403	11718	17456	17696	110	399966	0.01%	11.2/sec	24.19	7.16



Row Labels	StdDev of Latency	Max of Latency	Average of Latency
Book seat	19507.11781	394691	10884.17
Book ticket	22457.124	394854	11315.8
Cancel ticket	15985.19474	394067	10552.62
confirmation	13885.4865	392839	10458.71
Flights	16247.53672	394510	10553.66
Home Page	19440.08596	394893	10932.88
Itinerary Page	19413.23426	394358	10814.69
Itinerary Page 01	19465.22177	394505	10828.34
Itinerary Page menu	13922.77027	394489	10385.31
Login	20490.35854	394911	11007.62
re-researvation	16101.70714	394600	10645.38
Researvation	18050.17897	399962	10777.57
researvation book seat	20931.06232	394711	11028.66
researvation flight choose	17538.46074	394587	10676
Signout	20918.98212	394710	11015.4
Welcome	21101.56652	394724	11000.93
Grand Total	18711.25902	399962	10814.38

Observations:

1. **0.01%** errors encountered during load test around 42379 transactions.
2. Average Response time **10869 ms**
3. 95th percentile fall under – **17456 ms**
4. Throughput (requests per second) - **11.2/ sec**
5. Network performance (Max latency) – **399962 ms**
6. Network performance (Average latency) – **10814.38 ms**

Stress Test

In this stress test, which is conducted by stepping thread groups to perform 1000 users holds for 1200 seconds. In this load, 50 threads are gradually added by each 15-second interval.

Requests	Executions			Response Times (ms)							Throughput	Network (KB/Sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	169837	149750	88.17%	1660.34	0	349132	245.00	5822.60	6935.95	8713.00	100.26	252.54	7.59
Book seat	9480	8320	87.76%	1786.19	0	346457	198.00	7586.10	11385.70	18523.85	5.60	14.61	0.49
Book ticket	9410	8293	88.13%	1837.85	0	346501	194.00	6791.40	11250.90	18430.35	5.56	14.33	0.52
Cancel ticket	9299	8209	88.28%	1665.20	1	345627	196.00	6839.00	11250.00	18584.00	5.49	13.18	0.44
confirmation	9430	8249	87.48%	1648.13	1	345200	195.00	7752.50	11425.80	18592.80	5.57	14.46	0.64
Flights	9613	8477	88.18%	1585.73	0	345244	199.00	6706.60	11170.60	18525.18	5.68	14.01	0.35
Home Page	9457	8359	88.39%	1587.22	0	346295	193.00	6384.40	11243.00	18552.42	5.59	13.80	0.33
Itinerary Page	9326	8222	88.16%	1957.24	0	349132	195.00	6748.90	11173.60	18413.87	5.51	13.21	0.33
Itinerary Page 01	9449	8368	88.56%	1655.08	0	348290	194.00	6393.00	11201.50	18340.00	5.58	13.20	0.33
Itinerary Page menu	9365	8346	89.12%	1534.30	0	343910	192.00	4823.40	11109.80	18273.72	5.53	13.68	0.31
Login	19028	16723	87.89%	1656.83	0	345494	195.00	7258.20	11335.65	18494.39	11.23	26.81	0.82
re-researvation	9323	8240	88.38%	1580.88	0	346465	196.00	6656.60	11284.60	18551.60	5.51	15.32	0.39
researvation	9541	8393	87.97%	1687.27	0	346438	198.00	7283.80	11305.70	18430.28	5.64	14.58	0.61
Researvation	9378	8294	88.44%	1577.04	0	344250	195.00	6685.40	11285.10	18515.26	5.54	15.41	0.33
researvation book seat	9525	8422	88.42%	1494.86	0	345491	195.00	6560.40	11221.80	18453.22	5.63	14.68	0.49
researvation flight choose	9317	8202	88.03%	1651.10	0	345565	195.00	6898.00	11383.00	18481.20	5.51	14.95	0.53
Signout	9517	8380	88.05%	1748.60	0	346348	194.00	6803.40	11309.20	18446.64	5.63	13.42	0.34
Welcome	9379	8253	87.99%	1578.30	0	345687	195.00	6560.00	11119.00	18254.20	5.54	13.07	0.34

Row Labels	StdDev of Latency	Max of Latency	Average of Latency
Book seat	11676.25573	346407	1637.727
Book ticket	13184.17122	346467	1690.358
Cancel ticket	10636.26999	345615	1516.431
confirmation	8718.656655	345186	1501.559
Flights	9260.62636	345233	1435.928
Home Page	9372.192442	346291	1439.708
Itinerary Page	15008.25934	349125	1809.478
Itinerary Page 01	11058.77837	348277	1507.493
Itinerary Page menu	9893.33572	343905	1385.106
Login	9951.506642	345483	1509.938
re-researvation	9396.507343	346438	1431.698
Researvation	9987.557043	346383	1483.768
researvation book seat	7856.696126	345475	1345.897
researvation flight choose	10056.91197	345560	1502.893
Signout	11638.68781	346314	1601.706
Welcome	9341.75689	345683	1429.881
Grand Total	10509.36045	349125	1512.211

Observations:

1. During stress test error occurred which is **88.17%**.
2. Average Response time **166.40ms**
3. 95th percentile falls under – **6935.35ms**
4. Throughput (requests per second) – **100.26/s**
5. Network performance (Max latency) – **349125ms**
6. Network performance (Average latency) – **1512.211ms**

Spike Test

In this spike test, which is conducted by open model thread group to perform 1000 users interact within 40 seconds. In this test started by 20 seconds and completed by 80 seconds and maintained 1 minute of transaction rate up to 300. Load rate will be 300/seconds. Around 14330 samples were transitioned.

Requests	Executions			Response Times (ms)								Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent	
Total	14330	10824	75.53%	1184.40	1	23206	8.00	1470.50	11652.00	18050.35	85.10	239.16	13.28	
Book seat	785	606	77.20%	1034.72	1	18920	8.00	326.20	11174.30	17284.54	4.76	14.28	0.78	
Book ticket	797	613	76.91%	1210.56	2	19084	8.00	2500.20	11741.70	17866.12	4.95	14.55	0.91	
Cancel ticket	775	595	76.77%	1320.39	2	20462	8.00	3132.40	13737.80	18564.16	4.79	12.37	0.75	
confirmation	825	607	73.58%	1134.77	2	20618	9.00	3138.40	10844.50	17804.44	4.96	14.71	1.21	
Flights	832	622	74.76%	1159.32	1	19433	8.00	1443.90	11911.35	17882.53	4.94	13.31	0.64	
Home Page	807	591	73.23%	1300.70	2	19505	9.00	4746.20	11834.20	17792.64	4.88	13.05	0.67	
Itinerary Page	787	606	77.00%	1105.81	2	19622	8.00	520.40	11049.20	18414.04	4.92	12.74	0.57	
Itinerary Page 01	797	610	76.54%	1159.79	2	20773	8.00	1174.20	10756.60	18747.88	4.90	12.26	0.59	
Itinerary Page menu	785	605	77.07%	1238.47	1	19895	8.00	1466.40	12887.80	18078.66	4.76	12.97	0.57	
Login	1606	1213	75.53%	1203.47	2	20787	8.00	1687.60	11838.40	18008.55	9.81	24.98	1.44	
re-researvation	759	590	77.73%	1057.10	1	19122	8.00	707.00	10640.00	17433.80	4.54	15.13	0.61	
researvation	760	569	74.87%	1301.53	2	19579	8.00	3570.70	12109.55	18224.93	4.58	13.53	1.03	
Researvation	813	604	74.29%	1186.72	2	21213	8.00	1119.60	12484.20	18226.88	4.83	16.33	0.64	
researvation book seat	793	581	73.27%	1308.87	2	23206	8.00	3915.60	11545.40	18906.58	4.72	14.14	0.96	
researvation flight choose	767	579	75.49%	994.95	2	20456	8.00	991.00	10035.40	17883.84	4.64	14.89	0.92	
Signout	846	639	75.53%	1062.73	2	20273	8.00	852.80	10705.80	18241.65	5.21	13.18	0.65	
Welcome	796	594	74.62%	1338.66	1	20911	9.00	3466.60	13003.95	18438.24	4.81	11.85	0.62	

Row Labels	StdDev of Latency	Max of Latency	Average of Latency
Book seat	3478.934853	18901	1022.567
Book ticket	3771.015761	19024	1196.821
Cancel ticket	4120.997377	20433	1308.021
confirmation	3557.964177	20592	1122.481
Flights	3747.939195	19386	1147.788
Home Page	3882.236671	19485	1289.497
Itinerary Page	3734.870174	19605	1093.656
Itinerary Page 01	3826.067505	20721	1147.767
Itinerary Page menu	3935.596956	19831	1226.371
Login	3805.204194	20749	1191.166
re-researvation	3538.051772	19098	1043.161
Researvation	3950.7381	21162	1229.532
researvation book seat	4006.635601	23137	1296.629
researvation flight choose	3439.628065	20437	981.8305
Signout	3619.590046	20243	1052.33
Welcome	4124.373654	20880	1327.372
Grand Total	3798.463063	23137	1172.162

Observations:

1. During Spike test error occurred which is **75.53%**
2. Average Response time **1184.40ms**
3. 95th percentile Response falls under – **11652ms**
4. Throughput (requests per second) – **85.10/sec**
5. Network performance (Max latency) – **23137ms**
6. Network performance (Average latency) – **1172.162ms**

Endurance Test

In this endurance test, which is conducted by Ultimate thread group to perform 200 users interact within 62 minutes. In this test, the full load started after 60 seconds and was completed by 1 hour. Each request is integrated with a uniform random timer to simulate real-world scenarios. All the requests will under synchronize timer for 200 concurrent users.

Requests	Executions				Response Times (ms)						Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	40433	1	0.00%	10457.95	61	17482	10082.00	14927.80	16447.95	16636.99	10.91	23.56	6.96
Book seat	2298	0	0.00%	10484.54	83	17186	10097.50	15032.10	16466.05	16616.06	0.62	1.80	0.45
Book ticket	2225	0	0.00%	10545.44	71	17363	10102.00	15092.80	16477.70	16645.22	0.60	1.59	0.48
Cancel ticket	2260	0	0.00%	10448.95	80	17062	10096.50	15077.20	16465.90	16665.24	0.61	0.69	0.41
confirmation	2254	0	0.00%	10406.43	92	16975	10101.00	14942.50	16447.50	16659.60	0.61	1.71	0.56
Flights	2202	0	0.00%	10476.98	83	17304	10121.00	14946.40	16440.00	16635.67	0.60	1.02	0.31
Home Page	2307	0	0.00%	10384.80	72	17209	10068.00	14897.00	16444.60	16564.28	0.62	1.06	0.32
Itinerary Page	2231	0	0.00%	10506.15	75	17153	10107.00	15065.20	16459.00	16624.76	0.60	0.68	0.30
Itinerary Page 01	2156	1	0.05%	10382.49	61	17203	10102.00	14946.80	16424.05	16630.44	0.58	0.47	0.30
Itinerary Page menu	2298	0	0.00%	10486.96	78	17158	10072.00	15019.50	16459.15	16621.00	0.62	1.06	0.32
Login	4483	0	0.00%	10477.15	67	17203	10112.00	15007.20	16443.00	16617.32	1.21	1.29	0.73
re-reservation	2273	0	0.00%	10484.11	90	17033	10084.00	15069.00	16473.00	16657.26	0.61	2.71	0.37
reservation	2166	0	0.00%	10493.49	91	17482	10108.00	15007.30	16474.95	16698.96	0.59	1.60	0.52
Reservation	2300	0	0.00%	10497.79	106	17224	10134.00	14955.60	16473.00	16631.90	0.62	2.75	0.32
reservation book seat	2274	0	0.00%	10549.43	70	17310	10117.50	15036.00	16485.25	16673.25	0.61	1.80	0.47
reservation flight choose	2228	0	0.00%	10339.47	79	17015	10103.00	14841.00	16426.65	16617.84	0.60	2.27	0.49
Signout	2215	0	0.00%	10423.80	73	17442	10092.00	15047.40	16458.00	16627.28	0.60	0.60	0.31
Welcome	2263	0	0.00%	10375.54	65	16960	10071.00	14809.20	16416.60	16571.88	0.61	0.50	0.31

Row Labels	StdDev of Latency	Max of Latency	Average of Latency
Book seat	3098.516365	17176	10474.72
Book ticket	3178.032993	17294	10535.77
Cancel ticket	3246.048562	17028	10439.85
confirmation	3121.525814	16965	10396.79
Flights	3143.192041	17289	10468.22
Home Page	3089.116554	17177	10376.12
Itinerary Page	3233.114872	17147	10497.54
Itinerary Page 01	3250.901299	17198	10374.34
Itinerary Page menu	3143.738544	17143	10478.44
Login	3198.731969	17191	10468.47
re-reservation	3210.778207	17025	10473.45
reservation	3082.132028	17471	10485.08
reservation book seat	3108.326196	17287	10539.5
reservation flight choose	3116.786123	17007	10329.36
Signout	3165.89831	17427	10415.59
Welcome	3145.00934	16957	10367.13
Grand Total	3156.320879	17471	10448.68

Observations:

1. During Endurance test **Nil** Error present.
2. Average Response time **10457.96ms**
3. 95th percentile Response times – **16636.99ms**
4. Throughput (requests per second) – **10.91/Sec**
5. Network performance (Max latency) – **17471ms**
6. Network performance (Average latency) – **10448.68ms**

Volume Test

In this volume test, which is conducted by the Ultimate thread group to perform 800 users interact with 960 seconds. In this test, which was performed by various load actives 100, 500 and full load 800. This full load 800 users interact 6 minutes. Each request is integrated with a uniform random timer and random controller to simulate real-world scenarios.

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	52507	39778	75.76%	2368.55	0	19729	5.00	3759.80	6894.70	11630.89	55.30	136.39	8.54
Book seat	2936	2247	76.53%	2399.19	0	19616	4.00	11079.00	17398.30	17870.93	3.10	8.20	0.52
Book ticket	2919	2230	76.40%	2269.24	1	19644	4.00	10790.00	17234.00	17825.40	3.08	7.97	0.58
Cancel ticket	2803	2104	75.06%	2460.60	1	19693	4.00	11153.00	17387.80	17940.68	2.96	6.53	0.50
confirmation	2966	2234	75.32%	2401.04	1	19694	5.00	10974.60	17398.95	17968.31	3.13	8.21	0.71
Flights	3010	2248	74.68%	2435.95	0	19616	5.00	10912.70	17349.90	17943.78	3.18	7.47	0.42
Home Page	2873	2150	74.83%	2554.28	0	19689	5.00	11202.60	17454.30	17943.16	3.03	7.13	0.39
Itinerary Page	3040	2278	74.93%	2359.83	0	19633	4.00	10896.50	17315.95	17900.77	3.20	7.07	0.40
Itinerary Page 01	2900	2247	77.48%	2234.68	1	19592	4.00	10931.00	17308.50	18030.84	3.06	6.64	0.35
Itinerary Page menu	2822	2135	75.66%	2236.27	1	19568	4.00	10744.00	17173.35	17914.94	2.97	7.01	0.38
Login	5830	4424	75.88%	2312.69	1	19614	4.00	10913.00	17300.00	17934.14	6.14	13.56	0.89
re-reservation	2894	2205	76.19%	2404.77	1	19729	4.00	10946.50	17363.25	18031.65	3.05	9.18	0.44
reservation	2882	2199	76.30%	2259.27	1	19665	5.00	10853.80	17313.85	17888.04	3.04	7.91	0.64
Reservation	2942	2223	75.56%	2389.86	0	19585	5.00	11082.10	17306.40	17918.39	3.10	9.37	0.39
reservation book seat	2979	2219	74.49%	2555.34	1	19727	5.00	11178.00	17410.00	17973.80	3.14	8.36	0.61
reservation flight choose	2930	2268	77.41%	2154.49	1	19577	4.00	10777.80	17339.45	17880.38	3.09	8.76	0.56
Signout	2947	2258	76.62%	2366.11	1	19646	4.00	11065.00	17321.00	17875.20	3.11	6.85	0.37
Welcome	2834	2109	74.42%	2526.68	1	19556	4.00	11180.50	17379.00	17930.55	2.99	6.34	0.39

Type of error	Number of errors	% in errors	% in all samples
Non HTTP response code: org.apache.http.conn.HttpHostConnectException/Non HTTP response message: Connect to localhost:1080 [localhost:127.0.0.1, localhost/0.0.0.0:0.0.0.1] failed: Connection refused: connect	39778	100.00%	75.76%

Row Labels	StdDev of Latency	Max of Latency	Average of Latency
Book seat	5199.263336	19607	2392.907
Book ticket	4960.539791	19635	2262.598
Cancel ticket	5233.250513	19684	2454.718
confirmation	5164.711013	19678	2394.75
Flights	5125.564173	19612	2429.724
Home Page	5348.093618	19678	2548.172
Itinerary Page	5103.769178	19621	2354.009
Itinerary Page 01	5047.520177	19585	2228.798
Itinerary Page menu	4925.920339	19561	2230.152
Login	5054.499901	19609	2306.523
re-reservation	5166.401751	19721	2398.276
reservation	5071.010551	19659	2318.717
reservation book seat	5342.890986	19720	2549.078
reservation flight choose	4908.482096	19562	2148.186
Signout	5132.104213	19632	2360.275
Welcome	5301.731659	19549	2520.752
Grand Total	5124.68747	19721	2362.358

Observations:

1. During Volume test error occurred which is **75.76%**.
2. Average Response time **2368.55**
3. 95th percentile Response times – **6894.70ms**
4. Throughput (requests per second) – **55.30/s**
5. Network performance (Max latency) – **19721ms**
6. Network performance (Average latency) – **2362.358ms**

Sign off Stress Testing

In this Sign off Stress Testing, which is conducted by the Stepping thread group to perform 100 users interact with 600 seconds. In this test, which was performed by stepping up load activates 10 threads for each 30 seconds and maximum load of 100 threads reached by 5.15 minute. Each request is integrated with a random controller and Duration assertion (3000ms) to simulate real-world scenarios.

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label ^	#Samples ↕	FAIL ↕	Error % ↕	Average ↕	Min ↕	Max ↕	Median ↕	90th pct ↕	95th pct ↕	99th pct ↕	Transactions/s ↕	Received ↕	Sent ↕
Total	9912	3	0.03%	2701.26	94	8600	919.00	7656.00	7768.35	7903.74	10.67	10.69	5.44
Signout	9912	3	0.03%	2701.26	94	8600	919.00	7656.00	7768.35	7903.74	10.67	10.69	5.44

Type of error ↕	Number of errors ▼	% in errors ↕	% in all samples ↕
500/Internal Server Error	3	100.00%	0.03%

Observations:

1. During Sign off Stress Testing error occurred which is **0.03%**.
2. Average Response time **2701.26**
3. 95th percentile Response times – **7768.35ms**
4. Throughput (requests per second) – **10.67/s**
5. Network performance (Max latency) – **8597ms**
6. Network performance (Average latency) – **2693.78ms**

Flight Booking Stress Testing

In this Flight Booking Stress Testing test, which is conducted by the Stepping thread group to perform 500 users interact with 600 seconds. In this test, which was performed by stepping up load activates 30 threads for each 30 seconds and maximum load of 500 threads reached by 9.15 minute. Each request is integrated with a random controller and Duration assertion (3000ms) to simulate real-world scenarios.

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label ^	#Samples ↕	FAIL ↕	Error % ↕	Average ↕	Min ↕	Max ↕	Median ↕	90th pct ↕	95th pct ↕	99th pct ↕	Transactions/s ↕	Received ↕	Sent ↕
Total	55220	45054	81.59%	2013.73	1	19668	148.00	6092.80	8753.95	9965.99	44.05	117.98	5.88
Book seat	6221	5079	81.64%	2009.44	2	18897	109.00	9810.00	11536.90	17703.46	4.96	13.06	0.66
Book ticket	5986	4895	81.77%	1974.59	2	19164	108.00	9614.90	11295.30	17699.00	4.78	12.34	0.69
confirmation	6125	4967	81.09%	2096.14	1	19043	111.00	10374.20	11593.40	17856.74	4.89	12.77	0.85
Flights	5936	4841	81.55%	1996.87	1	18948	110.00	9610.00	11304.05	17729.89	4.74	11.42	0.45
re-researvation	6307	5135	81.42%	2070.58	1	18956	110.00	9931.60	11549.60	17884.04	5.03	14.66	0.57
researvation	6151	5036	81.87%	1985.77	1	19006	110.00	9629.00	11366.40	17679.88	4.96	12.89	0.80
Researvation	6210	5041	81.18%	2045.90	1	18919	110.00	9865.50	11423.80	17686.90	4.95	14.44	0.48
researvation book seat	6183	5108	82.61%	1922.53	2	18992	110.00	9585.60	11324.80	17785.64	4.94	12.98	0.65
researvation flight choose	6101	4952	81.17%	2019.27	1	19668	110.00	9698.40	11331.00	17710.92	4.87	13.60	0.74

Row Labels	StdDev of Latency	Max of Latency	Average of Latency
Book seat	4341.054664	18887	1925.958
Book ticket	4284.461252	19160	1889.24
confirmation	4467.679391	19037	2012.008
Flights	4303.557815	18939	1912.494
re-researvation	4441.74491	18939	1985.483
Researvation	4335.894872	18995	1931.636
researvation book seat	4268.543539	18980	1836.041
researvation flight choose	4308.132398	19662	1935.646
Grand Total	4343.989811	19662	1929.147

Observations:

1. During Flight Booking Stress Testing test error occurred which is **81.59%**.
2. Average Response time **2013.73**
3. 95th percentile Response times – **8753.95ms**
4. Throughput (requests per second) – **44.05/s**
5. Network performance (Max latency) – **19662ms**
6. Network performance (Average latency) – **1929.147ms**

Itinerary Booking Load Testing

In this volume test, which is conducted by the Concurrency thread group to perform 300 users interact with 1800 seconds. In this test, which was performed by stepping up load active 10 threads within 3 minutes it will reach maximum load of 300 threads reached by 2.45 minute. Each request is integrated with a random & throughput controller and Duration assertion (300ms) to simulate real-world scenarios.

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	59206	35118	59.31%	2774.22	0	18627	667.50	8878.90	9511.00	10692.99	29.92	60.07	6.74
Itinerary Cancel ticket	14914	8780	58.87%	2849.48	1	18627	129.00	9782.50	11289.25	14756.80	7.54	14.89	2.10
Itinerary Page	14727	8725	59.24%	2779.11	1	18592	122.00	9724.00	11216.20	14709.92	7.44	14.75	1.52
Itinerary Page 01	14906	8918	59.83%	2731.30	0	18601	118.00	9650.60	11231.20	14740.72	7.53	14.00	1.56
Itinerary Page menu	14659	8695	59.32%	2736.36	1	18524	121.00	9687.00	11035.00	14676.40	7.41	16.44	1.56

Type of error	Number of errors	% in errors	% in all samples
Non HTTP response code: org.apache.http.conn.HttpHostConnectException/Non HTTP response message: Connect to localhost:1080 [localhost/127.0.0.1, localhost/0:0:0:0:0:0:1] failed: Connection refused: connect	35115	99.99%	59.31%
500/Internal Server Error	3	0.01%	0.01%

Row Labels	StdDev of Latency	Max of Latency	Average of Latency
Itinerary Cancel ticket	4351.986027	18617	2803.709
Itinerary Page	4298.182212	18580	2732.778
Itinerary Page 01	4279.371695	18591	2685.24
Itinerary Page menu	4274.844332	18512	2691.064
Grand Total	4301.483288	18617	2728.349

Observations:

1. During Itinerary Booking Load Testing error occurred which is **59.31%**.
2. Average Response time **2774.22**
3. 95th percentile Response times – **9511ms**
4. Throughput (requests per second) – **29.92/s**
5. Network performance (Max latency) – **18617ms**
6. Network performance (Average latency) – **2728.349ms**

Conclusion

After conducting various tests from various thread groups, we will come to know

- **Load Test** - After performing load test we received **0.01%** of error around 42379 transactions.
- **Stress Test** - After performing stress test we received **88.17%** of error around 1.69lakh transactions.
- **Endurance Test** - After performing Endurance test, we received **0.00%** of error around 40k transactions.
- **Spike Test** - After performing Spike test, we received **75.53%** of error around 14k transactions.
- **Volume Test** - After performing the volume test, an error occurred, which is **75.76%** around 52507 transactions.
- **Flight Booking Stress Testing** - After performing Endurance test, we received **81.56%** of error around 55220 transactions.
- **Sign off Stress Testing** - After performing Spike test, we received **0.03%** of error around 10k transactions.
- **Itinerary Booking Load Testing** - After performing the volume test, an error occurred, which is **59.31%**.around 59202 transactions.

The website can handle endurance and load tests. This webpage is unable to handle stress tests, volume tests, and spike tests. In this webpage, the flight booking stress test creates more errors. Sudden peak, unable to handle. But this stress test is getting passed only on the on the signoff page.

Appendix

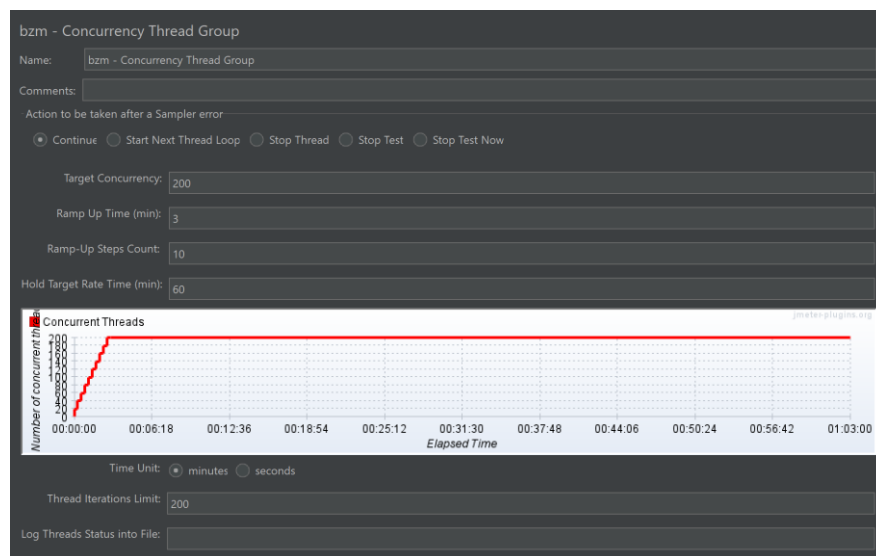
Load Test



Web-Tours - Load
Test.jmx

Configurations

- **Assertion** – No Assertion Integrated.
- **Timer** - Synchronizing Timer
- **Thread Group** - Concurrency Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



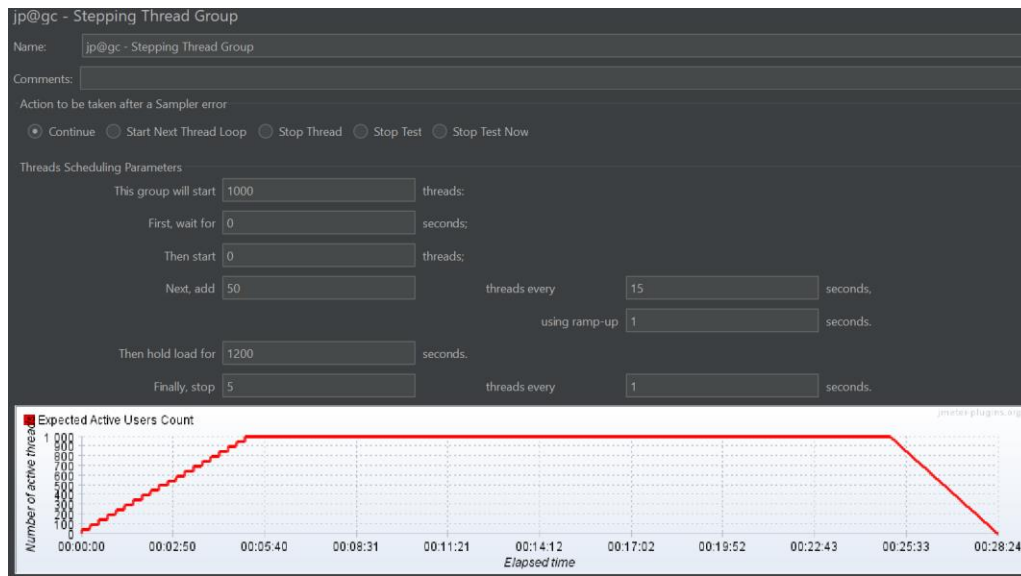
Stress Test



Web-Tours - Stress
Test.jmx

Configurations

- **Assertion** – No Assertion Added.
- **Timer** - Synchronizing Timer
- **Thread Group** - Stepping Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



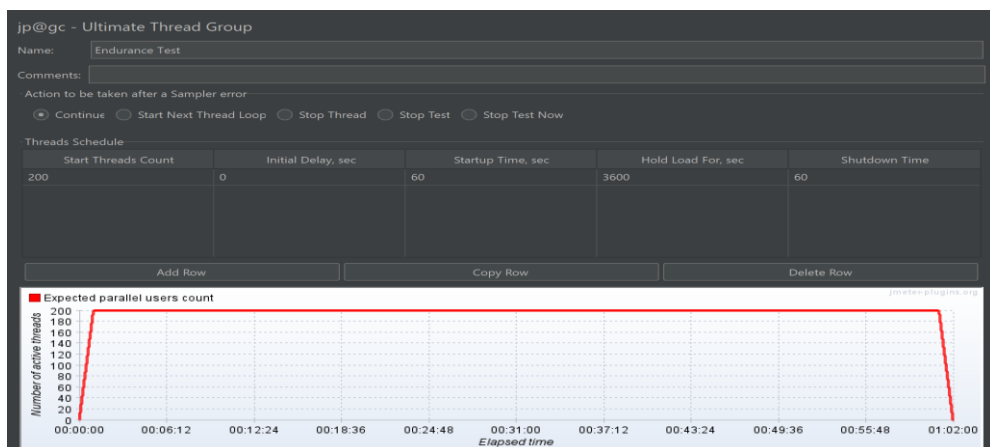
Endurance Test



Web-Tours -
Endurance Test.jmx

Configurations

- **Assertion** – Duration Assertion Integrated (3 Seconds)
- **Timer** - Synchronizing Timer, Uniform Random Timer
- **Thread Group** – Ultimate Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



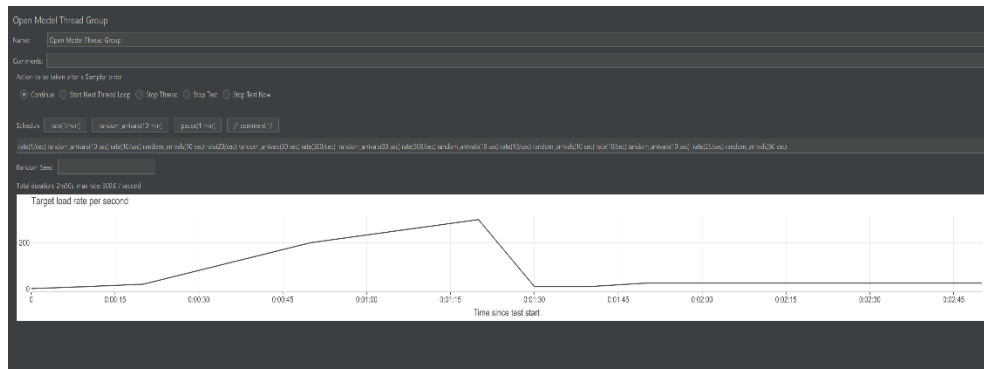
Spike Test



Web-Tours - Spike
Test.jmx

Configurations

- **Assertion** - No Assertion Integrated.
- **Timer** - Constant Throughput Timer
- **Thread Group** - Open Model Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager.



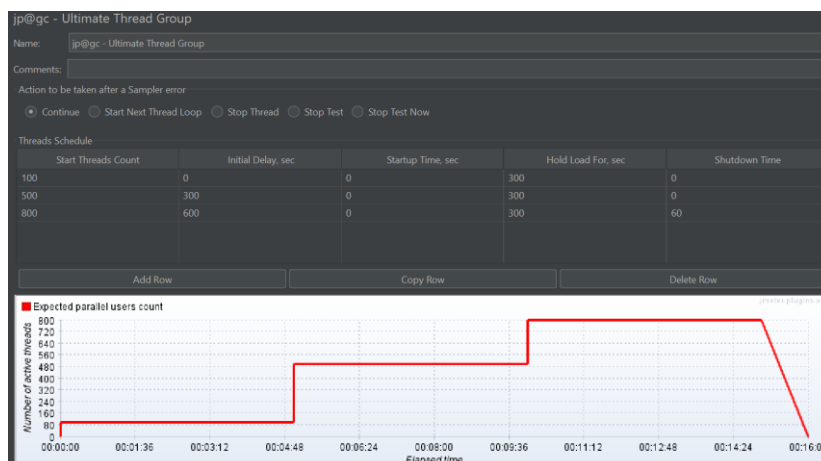
Volume Test



Web-Tours - Volume
Test.jmx

Configurations

- **Assertion** – No assertion added.
- **Timer** - Synchronizing Timer, Uniform Random Timer
- **Thread Group** - Ultimate Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



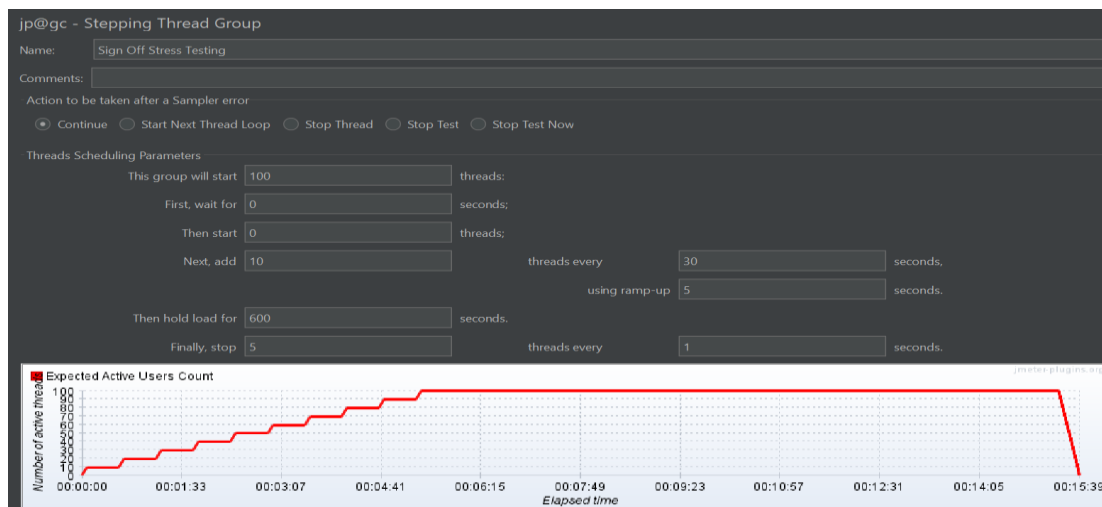
Sign off Stress Testing



Web-Tours - Sign Off
Stress Testing.jmx

Configurations

- **Assertion** – Duration assertion added.
- **Timer** - Synchronizing Timer
- **Thread Group** - Stepping Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



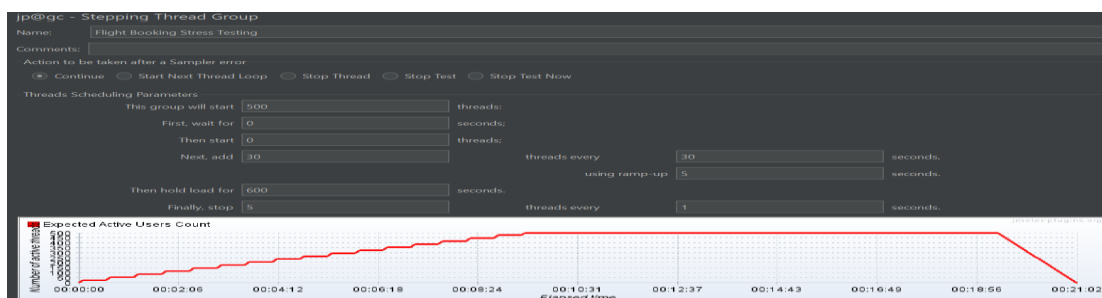
Flight Booking Stress Testing



Web-Tours - Flight
Booking Stress Testing.jmx

Configurations

- **Assertion** – Duration assertion added.
- **Timer** - Synchronizing Timer
- **Thread Group** - Stepping Thread Group
- **Controller** - Random Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager



Itinerary Booking Load Testing



Web-Tours - Itinerary
Management Load Te

Configurations

- **Assertion** – Duration assertion added.
- **Timer** - Synchronizing Timer
- **Thread Group** - Concurrency Thread Group
- **Controller** - Random Controller, Throughput Controller
- **Reports** - Summary, Aggregate, jp@gc graphs View Results Table & View Results Tree
- **Config Elements** - User Defined Variables, Header Manager, Cookie & Cache Manager

